SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1020	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	K. Yamuna	**Reg. No.:**	192511013
Date:	8-10-2026	**Duration:**	60 minutes

Code of Conduct

I K. Yamuna (192511013)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Yamuna K

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement**
   - Explain the problem in your own words.
   - Identify the objectives of the proposed system.
   - State the scope of the problem.
2. **Identify Key Stakeholders and Challenges**
   - Identify all the stakeholders involved in the system.
   - Explain the roles and responsibilities of each stakeholder.
   - Discuss the major challenges and issues faced by the stakeholders.
3. **Analyze the Current Approach**
   - Describe the existing system or current process.
   - Identify its strengths and limitations.
   - Analyze the problems or gaps that need to be addressed.
4. **Propose a Suitable Solution Using Software Engineering Principles**
   - Design a software-based solution for the given problem.
   - Describe the major functional modules of the proposed system.
   - Identify the functional and non-functional requirements.
   - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
5. **Justify the Chosen Methodology/Framework**

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. **Discuss Expected Outcomes and Limitations**
   - Describe the expected benefits of the proposed solution.
   - Explain how the solution addresses the identified challenges.
   - Discuss potential risks, implementation challenges, and limitations.
   - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification	Selects the most suitable SDLC/model/frame	Suitable methodology selected with	Methodology selected but justification is	Outcomes, limitations, or future

(10%)	work and provides strong technical justification.	adequate justification.	weak.	enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

CO1 AT1 – Case Study Analysis

Name: K. Yamuna

Reg no: 192511013

CSA 1019 – Software Engineering

PS- 38 Construction Project Progress and Material Tracking System

1. Understand the Problem Statement

Problem Explanation:

The Construction Project Progress and Material Tracking System is a software application used to monitor construction work and manage material availability. It reduces manual work, prevents material wastage, and helps complete projects on time.

Objectives of the Proposed System:

- To monitor construction progress efficiently.
- To track material stock and usage.
- To reduce material wastage and project costs.
- To improve communication among project members.
- To complete projects within the planned schedule.

Scope of the Problem:

The system manages project details, work progress, material inventory, supplier information, and report generation. It helps construction companies monitor their projects and resources in one place.

2. Identify Key Stakeholders and Challenges

Key Stakeholders and Their Roles:

- Project Manager: Plans activities and monitors project progress and budget.
- Site Supervisor: Updates daily work progress and checks material usage.
- Construction Workers: Complete assigned tasks and report work status.
- Storekeeper: Maintains material stock and usage records.
- Material Suppliers: Deliver materials according to project requirements.
- Clients: Monitor project progress and completion status.
- System Administrator: Manages user accounts and system security.

Major Challenges:

The major challenges include manual record errors, material shortages, delivery delays, budget overruns, material wastage, and poor communication between project members.

3. Analyze the Current Approach

Existing System:

The current system uses paper records, spreadsheets, phone calls, and site visits to track construction progress and material usage.

Strengths:

- Easy to understand and use.
- Requires less initial investment.
- Suitable for small construction projects.

Limitations:

- Manual records may contain errors.
- Progress updates may be delayed.
- Material stock details may not be accurate.
- Report preparation takes more time.

Problems or Gaps to Be Addressed:

The existing system lacks centralized data, automatic stock updates, and real-time progress monitoring. A digital solution is needed to improve accuracy, communication, and efficiency.

4. Propose a Suitable Solution Using Software Engineering Principles

Proposed Solution:

A web-based application is proposed to manage construction progress and material inventory. It stores project information in a centralized database and provides progress updates, stock details, reports, and alerts.

Major Functional Modules:

- User Management: Handles login and user access.
- Project Management: Maintains project details, budgets, and schedules.
- Progress Tracking: Records daily construction activities.
- Material Tracking: Monitors stock, purchases, and material usage.
- Supplier Management: Maintains supplier and delivery details.
- Reports and Alerts: Generates reports and sends alerts for low stock and project delays.

Functional Requirements:

The system should support secure login, project creation, progress updates, material stock management, supplier records, and report generation.

Non-Functional Requirements:

The system provide good performance, reliability, usability, scalability, and easy maintenance.

Application of Software Engineering Principles:

- Modularity: Divides the system into separate functional modules.
- Scalability: Supports additional projects and users.
- Maintainability: Allows easy updates and modifications.
- Reliability: Ensures accurate data storage and retrieval.
- Usability: Provides a simple and user-friendly interface.
- Security: Protects data through authentication and access control.

5. Justify the Chosen Methodology/Framework

Selected Methodology: Agile Model

The Agile model is selected because it supports step-by-step development, continuous testing, and regular feedback from users.

Reasons for Choosing Agile:

- Development can be divided into smaller stages.
- Each module can be tested separately.
- User feedback can be included regularly.
- Changes in requirements can be handled easily.
- Errors can be identified and corrected early.

Support for Project Development and Maintenance:

Agile helps develop the system module by module, starting with basic features and gradually adding advanced functions. Regular testing and feedback improve software quality and simplify future maintenance.

6. Discuss Expected Outcomes and Limitations

Expected Benefits:

- Improved construction progress monitoring.
- Accurate material stock management.
- Reduced material wastage and expenses.
- Better communication among stakeholders.
- Faster report generation and decision-making.
- Improved control over project schedules.

How the Solution Addresses the Challenges:

The system centralizes project information, tracks material usage, and provides timely progress updates. Alerts help identify material shortages and project delays, enabling managers to take quick action.

Potential Risks and Limitations:

- Initial development costs may be high.
- Employees may require training.
- Internet connectivity may affect system access.
- Incorrect data entry may lead to inaccurate reports.
- Security issues may arise without proper protection.

Future Enhancements:

- Develop a mobile application for site updates.
- Use QR codes or RFID for material tracking.
- Apply AI to predict project delays and material requirements.
- Integrate billing and accounting systems.
- Use IoT sensors to monitor equipment and site conditions.

Conclusion

The Construction Project Progress and Material Tracking System improves construction management by reducing manual work, tracking materials, and monitoring project progress. Using software engineering principles and the Agile model helps create a reliable, flexible, and maintainable system. Future enhancements can further improve construction efficiency.